

Dataset Generation Report

Extractive Distillation ML Digital Twin - Software Used

1. DWSIM 8.8
2. ChemSep
3. Python
4. Microsoft Excel

1. Objective

The objective of this stage was to generate a high-quality engineering dataset that could be used for developing machine learning models capable of predicting the performance of an industrial extractive distillation process.

Instead of relying on publicly available datasets, every data point was generated through rigorous process simulation using DWSIM and ChemSep. This approach ensured that all samples obeyed the governing principles of mass transfer, thermodynamics, phase equilibrium and process engineering.

The generated dataset forms the foundation of the Digital Twin developed in this project.

2. Motivation

The published research paper successfully demonstrated the feasibility of ethanol dehydration using extractive distillation. However, it did not provide a complete simulation dataset suitable for machine learning applications.

Several important operating parameters required for digital twin development were either absent or insufficiently explored. Consequently, a systematic simulation framework was developed to generate reliable process data over a broad operating region.

Rather than reproducing a single operating condition, the objective was to capture the behaviour of the process under multiple operating scenarios, thereby creating a dataset capable of supporting predictive modelling and process optimization.

3. Methodology

Dataset generation followed a sequential engineering workflow.

Initially, the extractive distillation process was reconstructed from the published literature. Missing operating conditions were identified and completed using engineering judgement together with systematic convergence studies.

Sensitivity analyses were then carried out for both distillation columns to determine the influence of major operating variables.

Following validation of the simulation model, additional operating conditions were investigated by varying multiple process variables over physically meaningful ranges.

Each simulation was manually executed in DWSIM, evaluated for convergence and checked for physical consistency before inclusion in the final dataset.

This workflow ensured that every sample represented a feasible industrial operating condition.

4. Source of Data

Source	Description
Research Paper	Initial process configuration
DWSIM	Dynamic process simulation
ChemSep	Rigorous distillation calculations
Sensitivity Analysis	Determination of operating ranges
Validation Study	Selection of reliable reference case
Combined Simulation Studies	Dataset generation

5. Operating Variables

The final dataset includes the five operating variables identified as having the greatest influence on process performance.

Input Variable	Unit	Description
Column 1 Boilup Ratio	–	Vapor generation rate in extractive column
Column 2 Bottom Flow	kmol/s	Bottom product specification
Column 2 Reflux Ratio	–	Solvent recovery column reflux ratio
Column 2 Pressure	bar	Operating pressure
Solvent Feed Temperature	°C	Ethylene glycol feed temperature

6. Process Responses

For every operating condition, the following engineering responses were recorded.

Output Variable	Unit
Ethanol Purity	wt%
Ethanol Recovery	%
Ethylene Glycol Recovery	%
Total Energy Consumption	kW
Process Performance Index (PPI)	–

7. Dataset Statistics

Property	Value
Total Simulations Performed	86
Converged Cases Used	73
Input Variables	5

Output Variables	5
Simulation Software	DWSIM
Thermodynamic Model	NRTL
Solvent	Ethylene Glycol

8. Engineering Constraints

Only physically meaningful simulations were retained.

The following criteria were enforced throughout dataset generation.

- Only fully converged simulations were accepted.
- Material and energy balances had to converge successfully.
- Unrealistic temperatures were discarded.
- Non-physical compositions were removed.
- Operating conditions causing instability were excluded.
- The validated operating condition served as the reference for subsequent studies.

These constraints ensured that the machine learning model would learn realistic industrial behaviour rather than numerical artifacts.

9. Parameter Coverage

The operating ranges investigated during this study are summarized below.

Variable	Minimum	Maximum
Column 1 Boilup Ratio	0.35	0.59
Column 2 Bottom Flow	0.06 kmol/s	0.09 kmol/s
Column 2 Reflux Ratio	0.50	3.00
Column 2 Pressure	0.8 bar	2.0 bar
Solvent Feed Temperature	180°C	256°C

These ranges were selected based on sensitivity analysis and convergence studies rather than arbitrary values.

10. Process Performance Index (PPI)

To compare operating conditions using multiple engineering objectives simultaneously, a Process Performance Index (PPI) was developed.

The PPI combines

- Ethanol Purity
- Average Recovery (Ethanol + Ethylene Glycol)
- Specific Energy Consumption

Each parameter was normalized before combination into a single objective function.

Higher PPI values indicate operating conditions with superior purity, higher recovery and lower energy consumption.

This metric was subsequently used to rank all operating conditions and train the machine learning models.

12. Dataset Quality Assessment

The generated dataset possesses several characteristics desirable for machine learning.

- Generated using rigorous first-principles process simulation.
- Covers a broad operating window.
- Includes both optimal and sub-optimal operating conditions.
- Contains nonlinear process behaviour.
- Includes realistic industrial operating constraints.
- Suitable for regression, optimization and digital twin development.

Unlike purely statistical datasets, every sample satisfies thermodynamic consistency and process engineering principles.

13. Correlation Analysis

A correlation analysis was performed to understand the relationships between process inputs and engineering responses before machine learning.

The resulting correlation heatmap provides insight into

- variables with the strongest influence on ethanol purity,
- operating parameters affecting energy consumption,
- interactions among process variables,
- independent variables with minimal correlation.

This analysis also serves as an important preliminary feature engineering step.

14. Final Dataset

The final dataset consists of 73 validated operating conditions generated through systematic simulation studies.

Each row represents one complete extractive distillation simulation, including both operating conditions and resulting process performance indicators.

This dataset forms the input for the machine learning models developed in the subsequent stage of the project.

15. Conclusion

A complete engineering dataset was successfully generated from validated DWSIM simulations using a systematic workflow involving process development, sensitivity analysis, validation and optimization studies.

Unlike datasets obtained from experimental databases or random sampling, every observation was derived from rigorous first-principles simulation and verified for engineering feasibility.

The resulting dataset provides a reliable foundation for machine learning model development, digital twin construction and future AI-assisted optimization of extractive distillation processes.